

BISHOPS HEAD, MD, USA

WMO: 997688

Lat: 38.220N

Lon: 76.039W

Elev: 3

StdP: 101.29

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.3	(c) -4.9	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.7	(k) -1.8	(l) 13.7	(m) -2.7	(n) 6.1	(o) 330	(p) 0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 4.5	(c) 31.2	(d) N/A	(e) 30.1	(f) N/A	(g) 29.2	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 4.5	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 12.4	(b) 11.0	(c) 9.4	(e) -8.7	(f) 33.0	(g) 1.6	(h) 1.5	(i) -9.9	(j) 34.1	(k) -10.9	(l) 34.9	(m) -11.8	(n) 35.8	(o) -12.9	(p) 36.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	2.3	3.4	7.4	13.4	19.5	24.2	26.4	25.5	22.1	16.5	9.6	5.4		
	DBStd	9.22	4.41	4.15	4.36	3.78	3.46	2.73	2.13	2.07	2.79	3.76	3.44	3.58		
	HDD10.0	742	242	187	105	12	0	0	0	0	0	2	47	147		
	HDD18.3	2183	498	418	341	151	29	1	0	0	3	79	261	401		
	CDD10.0	2461	2	2	23	116	293	427	509	481	364	204	36	4		
	CDD18.3	859	0	0	0	4	64	178	251	223	117	22	0	0		
	CDH23.3	6195	0	0	0	4	181	1234	2367	1838	540	31	0	0		
	CDH26.7	1283	0	0	0	0	17	236	622	356	52	1	0	0		
(14) Wind	WSAvg	4.8	4.8	4.9	5.2	5.2	5.1	4.9	4.6	4.4	4.5	5.0	4.8	4.7		
(15) Precipitation	PrecAvg	1168	82	72	99	90	92	112	115	126	106	100	77	93		
	PrecMax	1667	183	187	258	199	224	264	262	258	274	204	198	202		
	PrecMin	899	38	13	1	48	29	40	38	37	22	17	9	39		
	PrecStd	175	33	39	48	32	50	53	53	52	65	55	48	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.6	14.5	18.7	23.8	28.7	31.6	33.0	32.2	29.3	25.8	18.5	15.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	12.3	12.1	16.7	21.5	26.6	30.0	31.7	30.4	28.0	24.1	16.7	13.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	10.4	10.6	15.0	19.9	25.3	29.0	30.5	29.4	27.2	23.1	15.4	12.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	8.3	9.0	13.4	18.5	24.0	28.0	29.6	28.5	26.3	21.7	14.5	10.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.5	5.8	5.9	5.5	5.1	4.5	4.5	4.7	4.8	5.1	5.0		
		MCDBR	6.8	7.1	7.4	6.6	5.9	5.6	5.0	4.7	4.8	5.0	5.8	6.5		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.320	0.354	0.393	0.437	0.484	0.514	0.506	0.408	0.360	0.327	0.314		
	taud		2.504	2.479	2.425	2.332	2.243	2.162	2.089	2.099	2.350	2.485	2.533	2.540		
	Ebn at Noon		871	911	907	885	849	806	777	773	838	852	847	845		
	Edh at Noon		79	91	106	124	138	150	160	155	114	90	76	71		
(44) All-Sky Solar Radiation	RadAvg		2.20	3.02	4.05	5.13	5.82	6.38	6.17	5.53	4.49	3.42	2.49	1.94		
	RadStd		0.18	0.32	0.34	0.36	0.62	0.47	0.29	0.32	0.44	0.36	0.21	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)		+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+91

Nomenclature: See separate page